4.6.1 Technological challenges

One of the key challenges of third-generation drones was the dependency on satellite communication (SATCOM) and ground control stations for navigation and mission management. This reliance made them vulnerable to signal interference, jamming, and cyberattacks, particularly in contested or GPS-denied environments. Although many drones were capable of semi-autonomous flight, full autonomy was still restricted by limited onboard computing power and sensor fusion capabilities. Furthermore, the integration of multi-sensor systems, including EO/IR, SAR, and ELINT, posed challenges in terms of data synchronization and processing latency. Maintenance of advanced composite airframes, stealth coatings, and precision payload systems also required high technical expertise, increasing operational costs.

4.6.2 Operational limitations

Operationally, third-generation UAVs faced constraints in mission flexibility and endurance under combat conditions. While systems like the MQ-9 Reaper and RQ-4 Global Hawk achieved extended flight durations, their large size and radar signatures made them susceptible to detection in hostile airspace. Harsh environmental conditions such as strong winds, sandstorms, or high-altitude icing also affected flight stability and data accuracy. Additionally, airspace integration with civilian aircraft remained a significant challenge, as regulatory frameworks for unmanned operations were still evolving globally.

4.6.3 Strategic and logistical constraints

Strategically, the widespread use of UAVs raised concerns regarding data security, export control, and international compliance. Many third-generation drones, particularly those equipped with strike capabilities, were